

Multiple Choice (10 marks)

1. How many distinct diagonals of a convex hexagon can be drawn?
 - (a) 6
 - (b) 8
 - (c) 9
 - (d) 36
2. Find the value of the expression $14 + (-71)$.
 - (a) -85
 - (b) 57
 - (c) 85
 - (d) -57
3. The original price of a new bicycle is \$138.00. If the bicycle is marked down 15%, what is the new price?
 - (a) \$20.70
 - (b) \$117.30
 - (c) \$123.00
 - (d) \$153.00
4. Suppose a , b , and c are positive numbers satisfying $a^2/b = 1$, $b^2/c = 2$, $c^2/a = 3$. Find a .
 - (a) $12^{(1/7)}$
 - (b) $7^{(1/12)}$
 - (c) 1
 - (d) 6
5. Roya paid \$48 for 12 cartons of orange juice. What is the unit rate per carton of orange juice that Roya paid?
 - (a) \$3
 - (b) \$4
 - (c) \$6
 - (d) \$12

True / False (10 marks)

Answer True or False

6. True or False: The number 300 is a powerful integer since it contains prime factors with their squares as factors.
7. True or False: The maximum value of the function $f(x) = xe^{\frac{1}{x}} - x$ is e .
7. True or False: The quotient of the expression 2,314 divided by 4 is 578 with a remainder of 2.
8. True or False: The solution to the equation $-47 = g + 24$ is $g = -23$.
9. True or False: The value of the expression 28×42 is 1,176.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how many distinct arithmetic sequences of consecutive odd integers can be formed that sum to 240. Provide a detailed explanation of your reasoning and calculations.
12. Calculate the number of different committees of 5 people that can be formed from a group of 15. Discuss the mathematical principles involved in your solution.

End of Paper